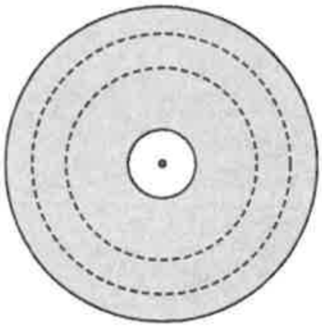
Unit 2.5 Mensuration

Paper 1: Intermediate

1.



The diagram shows four circles, all with the same centre.

The radius of the innermost circle is 5 cm and the radius of the outermost circle is 23 cm.

The shaded region is divided into three equal areas by the two dotted circles.

Find the radius of the larger dotted circle. [3]

[N22/I/10]

2. The volume of a cuboid is 1200 cm^3.

The area of the **largest** face is 150 cm^2.

The dimensions of the cuboid have integer values.

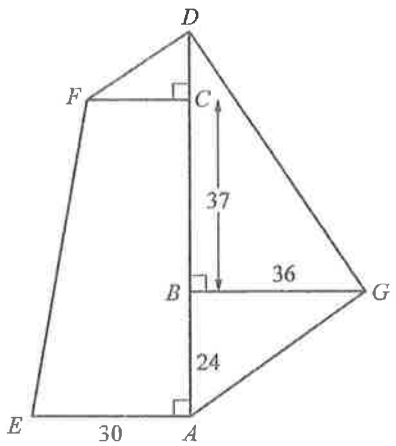
Find the dimensions of the cuboid. [2]

[N21/I/7]

3. The diagram shows a sketch of a horizontal field, *AEFDG*.

A survey of the field was carried out and measurements taken.

The measurements, in metres, of some of the lengths are shown on the diagram.



Angles *BAE*, *DCF* and *DBG* are right angles.

ABCD is a straight line.

(a) Angle *BGD* = angle *BAG*.

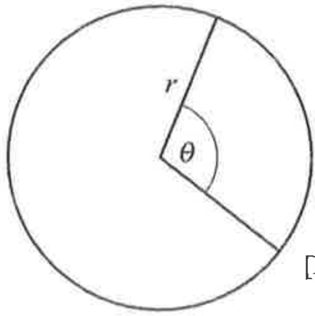
Show that *BD* = 54 m. [1]

(b) The area of section *ACFE* = 1586 m^2.

Calculate the total area of the field *AEFDG*. [4]

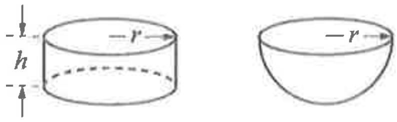
[N20/I/22]

4. The diagram shows a circle with radius r cm.
The circle is divided into two sectors.
The angle of the minor sector is θ radians.
The perimeter of the major sector is twice the perimeter of the minor sector.
Find the value of θ.
Give your answer correct to three decimal places.



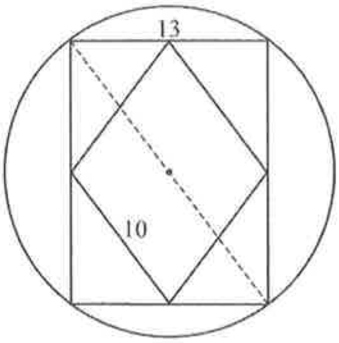
[4]
[N18/I/16]

5. A solid cylinder has radius r cm and height h cm.
A solid hemisphere has radius r cm.
The volumes of the cylinder and hemisphere are equal.
Work out, in terms of r, the total surface area of the cylinder.



[3]
[N17/I/9]

6. The diagram shows a rhombus drawn inside a rectangle inside a circle.



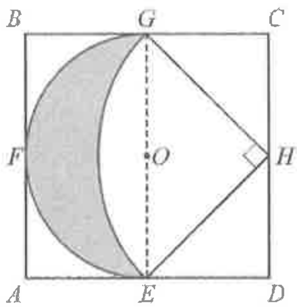
The rhombus has sides of length 10 cm.
The length of the shorter side of the rectangle is 13 cm.

(a) Calculate the circumference of the circle. [2]

(b) Calculate the area of the rectangle. [2]

[N16/I/16]

7.



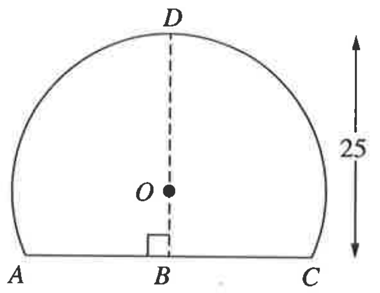
$ABCD$ is a square, centre O.
$BC = 2r$.
E, F, G and H are the midpoints of the sides of the square.
EFG is a semi-circular arc, centre O.
The other arc EG has centre H.
$HG = \sqrt{2}r$.
What fraction of the square $ABCD$ is **not** shaded? [5]

[N16/I/24]

8. $ABCD$ is a major segment of a circle, centre O and radius 17 cm.

$BD = 25$ cm.

Angle $ABD = 90°$.



Calculate the area of the segment. [5]

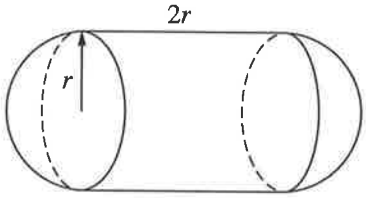
[N15/I/19]

9. **(a)** The surface area of a solid is given by $A = \pi p(2p + q)$.

Make q the subject of the formula. [2]

(b) Another solid is made from a cylinder and two hemispheres.

The cylinder has radius r and length $2r$ and the hemispheres have radius r.



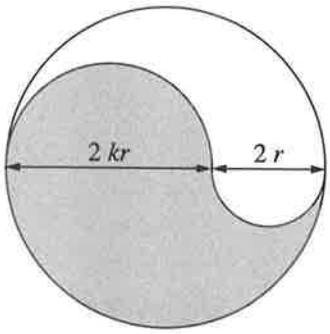
The total surface area of the solid is twice the total surface area of a cone with radius r and slant height l.

Find l in terms of r. [3]

[N15/I/21]

10. This design is drawn using a large circle and semicircles.

The diameters, in centimetres, of two of the semicircles are shown.



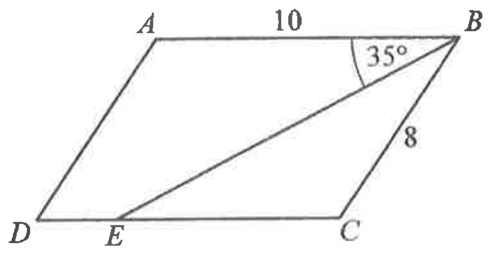
(a) Show that the total area, A, of the large circle is given by the formula $A = \pi r^2(k + 1)^2$. [2]

(b) Find, in terms of π and r, the difference in area between the shaded section and the unshaded section when $k = 2$. [4]

[N15/I/23]

Paper 1: Advanced

11.



ABCD is a parallelogram.

BE bisects angle *ABC*.

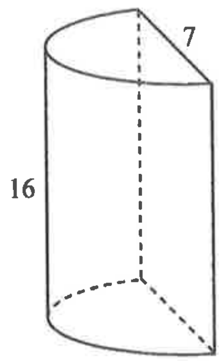
$AB = 10$ cm, $BC = 8$ cm and angle $ABE = 35°$.

Calculate the area of trapezium *ABED*. [5]

[N18/I/22]

Paper 2: Basic

1.



The diagram shows a solid in the shape of a half cylinder.

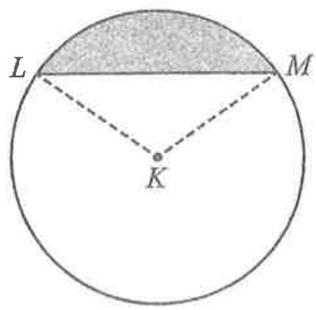
The diameter is 7 cm and the height is 16 cm.

(i) Calculate the volume of the solid. [2]

(ii) Calculate the total surface area of the solid. [3]

[N22/II/5(a)]

2.



The diagram shows a circle, centre *K*, radius 12 cm.

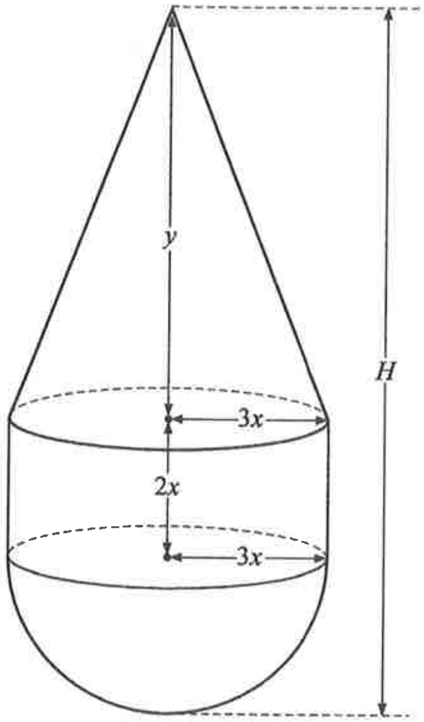
Angle $LKM = 1.8$ radians.

(i) Calculate the length of the major arc *LM*. [2]

(ii) Calculate the percentage of the circle that is shaded. [4]

[N20/II/6(b)]

3.



The diagram shows a solid formed from a cone, a cylinder and a hemisphere.

The cone has base radius $3x$ cm and height y cm.

The cylinder has radius $3x$ cm and height $2x$ cm.

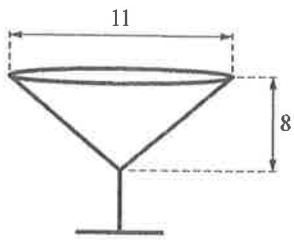
The hemisphere has radius $3x$ cm.

(a) The volume of the cone is twice the volume of the hemisphere.
Show that $y = 12x$. [3]

(b) The total surface area of the solid is 300 cm^2.
Calculate the total height, H, of the solid. [6]

[N21/II/4]

4.



A glass is in the shape of a cone on a stem.

The diameter of the top of the glass is 11 cm.

The height of the cone is 8 cm.

The thickness of the glass is negligible.

(a) Calculate the curved surface area of the inside of the glass. [3]

(b)

Amir pours water into the glass.

The surface of the water is 2 cm below the top of the glass.

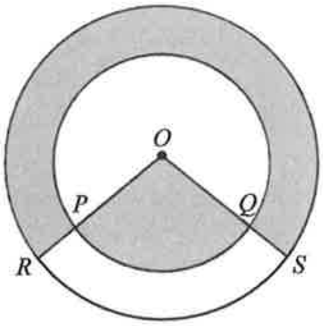
(i) Amir thinks that the glass is filled to 75% of its total capacity.
Explain why he is wrong. [1]

(ii) Calculate the percentage of the total capacity of the glass that is filled. [2]

(iii) Amir pours the water from this glass into a second glass.
The second glass is in the shape of a cylinder.
The depth of water in the cylindrical glass is 2.5 cm.
Calculate the radius of the cylindrical glass. [4]

[N20/II/5]

5.



P and *Q* are points on the circle centre *O* with radius 4 cm.

R and *S* are points on the circle centre *O* with radius 6 cm.

OPR and *OQS* are straight lines.

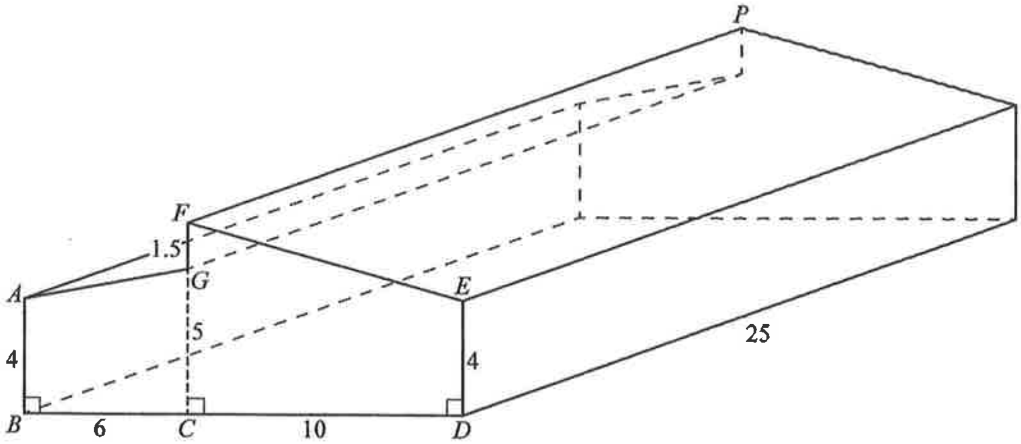
The perimeter of the minor sector *OPQ* is 15.2 cm.

(i) Calculate angle *POQ* in radians. [2]

(ii) Calculate the total shaded area. [3]

[N19/II/6(b)]

6.



The diagram shows a barn in the shape of a prism of length 25 m with a rectangular base.

The barn has two sloping rectangular roofs.

$AB = DE = 4$ m, $BC = 6$ m, $CD = 10$ m, $CG = 5$ m and $FG = 1.5$ m.

The barn is positioned on horizontal ground and the walls are vertical.

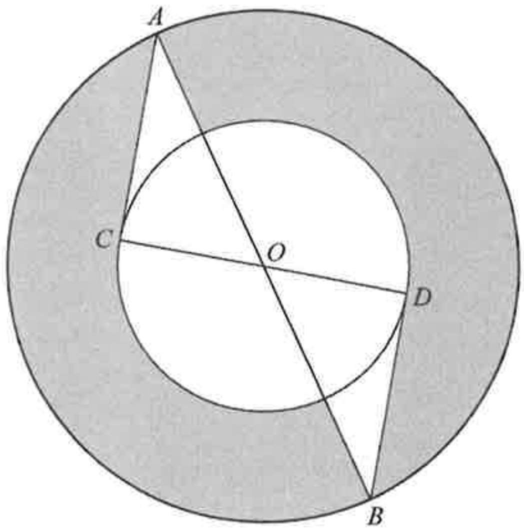
(a) Calculate the volume of the barn. [3]

(b) Calculate the total area of the two sloping roofs of the barn. [4]

(c) Calculate the angle of elevation of P from D. [4]

[N19/II/8]

7.



AB is a diameter of the large circle, centre O.

CD is a diameter of the small circle, centre O.

AC and BD are tangents to the small circle.

(a) Show that triangle OAC is congruent to triangle OBD.

Give a reason for each statement you make. [3]

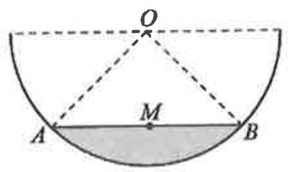
(b) The radius of the large circle is 7 cm and $O\hat{A}C = 30°$.

(i) Calculate the area of triangle OAC. [3]

(ii) Calculate the shaded area. [3]

[N18/II/7]

8.



The diagram shows a semicircle, centre O, radius 30 cm.

M is the midpoint of the chord AB.

$OM = 20$ cm.

(a) Show that angle $AOB = 96.4°$, correct to 3 significant figures. [2]

(b) Calculate the shaded area. [4]

(c) The semicircle is the cross section of a water trough of length 1.5 m, standing on level ground.

The shaded area represents the water in the trough.

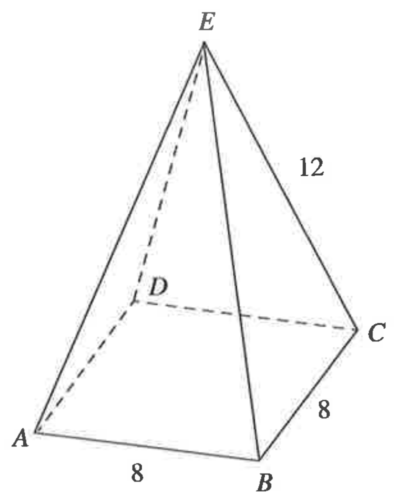
(i) Calculate the volume of water, in cm^3, in the trough. [2]

(ii) Calculate the number of litres of water that must be added to fill the trough.

Give your answer correct to the nearest 10 litres. [3]

[N17/II/8]

9.



The diagram shows a candle in the shape of a pyramid $ABCDE$.

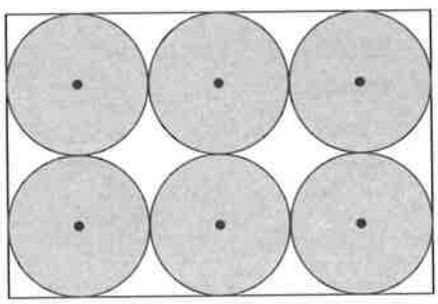
$ABCD$ is a square of side 8 cm and $AE = BE = CE = DE = 12$ cm.

(a) Calculate the volume of the candle. [4]

Another candle is made in the shape of a sphere.

The volume of this candle is the same as the volume of candle $ABCDE$.

(b) Show that the radius of the spherical candle is 3.78 cm, correct to 3 significant figures. [2]



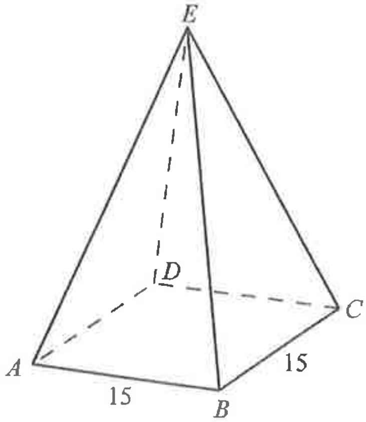
The diagram shows the plan view of a box holding six of the spherical candles.

The box is in the shape of a cuboid and the candles just fit into the box.

(c) Calculate the volume of empty space in the box. [3]

[N15/II/7]

10.



The diagram shows a pyramid $ABCDE$.

The base of the pyramid is a square of side 15 cm.

E is vertically above the centre of the square base.

The vertical height of the pyramid is 20 cm.

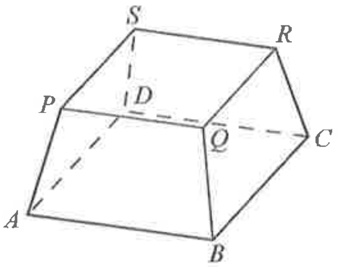
(a) Show that $AE = 22.6$ cm, correct to three significant figures. [3]

(b) Calculate angle BAE. [3]

(c) Calculate the total surface area of the pyramid. [3]

(d) A smaller, similar pyramid is removed from the top of the original pyramid.

The diagram below shows the solid remaining.

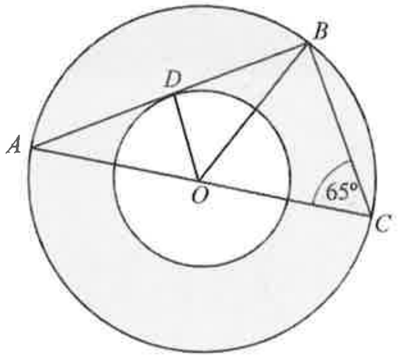


The vertical height of the solid remaining is 4 cm.

Calculate the area of the top, $PQRS$, of the remaining solid. [2]

[N18/II/8]

11.



The diagram shows two concentric circles, centre O.

A, B and C are points on the larger circle and D is a point on the smaller circle.

ADB is a tangent to the smaller circle and angle $BCO = 65°$.

(a) Show that triangles ABC and ADO are similar.

Give a reason for each statement you make. [2]

(b) Find the ratio area of triangle BOC : area of triangle ADO. [2]

(c) The radius of the smaller circle is 3 cm.

Find the shaded area. [4]

[N16/II/6]